SE444: Introduction to Artificial Intelligence

Makeup Mid-Term II — Answer Sheet

Fall 2025

Date: Wednesday, December 10, 2025

Student Information

Student ID:	Student Name:

Instructions:

- Write your answers clearly in the provided spaces.
- Show all work for partial credit.
- Circle your answer(s) for multiple choice questions.

Part I. Multiple Choice (4 pts, 20 min)

Instructions: Mark **X** for correct answer(s). Each question may have 1 or more correct answers.

Option	Q1 (5m)	Q2 (5m)	Q3 (5m)	Q4 (5m)
A				
B				
C				
D				
E				

Part II. Logical Reasoning (10 pts, 40 min)

Question II.1: Translation to FOL (3 pts, 10 min)

Statement: *"Every patient who is treated by a doctor at a hospital is prescribed at least one medication that treats one of their diseases."*

Answer:

Question II.2: Translation with Negation (3 pts, 10 min)

Statement: *"No doctor prescribes medication to a patient they have not examined."*

Answer:

Question II.3: Negation and CNF Conversion (4 pts, 20 min)

Negate and convert to **CNF** (show all steps including Skolemization):

$$\forall x \, (\text{Doctor}(x) \rightarrow \exists y \, (\text{Patient}(y) \wedge \text{TreatedBy}(y, x)))$$

Answer:

Part III. CSP (6 pts, 30 min)

Context: AI Conference scheduling with 3 rooms and 3 speakers.
Rooms: Room 1 (10:00–11:00), Room 2 (10:30–11:30), Room 3 (11:00–12:00)
Speakers: Dr. Smith (Rooms 1,2), Dr. Jones (Rooms 2,3), Dr. Brown (Rooms 1,2,3)

Question III.1: Formulate as CSP (2 pts, 10 min)

Define **variables**, **domains**, and **binary constraints**.

Answer:
Variables:

Domains:

Constraints:

Question III.2: Apply AC-3 Algorithm (3 pts, 15 min)

Show **sequence of arc revisions** and domain updates until fixed point.
For each revision: arc processed, values removed, updated domain.

Answer:

Question III.3: Find Consistent Solution (1 pt, 5 min)

Provide **one valid assignment** and explain why it satisfies all constraints.

Answer: